

Defending Deep Learning Based Network Intrusion Detection Systems Against Backdoor Attacks Using Machine Unlearning

Daniel Alvarado

Raj Patel

Varkey M. John

Abstract

In this paper, we develop a novel machine unlearning technique to remove backdoor samples from poisoned machine learning based network intrusion detection systems (ML-IDS). This is accomplished by using a rule-based IDS called ‘Suricata’ to clean the poisoned training dataset, which in turn is used to fine-tune the ML-IDS. This enables the unlearning, thus effectively removing the poisoned samples. With our approach, the un-poisoned ML-IDS could correctly classify 100% of the poisoned samples as ‘attack’. The application of our method is very relevant today, as more IDS systems are switching to use ML to improve their detection capabilities.

1 Introduction

Network intrusion detection systems (IDSs) have evolved to use deep learning techniques to effectively detect malicious intruders. However, adversaries can add backdoors to these deep learning models by training them with certain trigger patterns. When such a model is used in testing, if a trigger is present in the input, the IDS can be manipulated into overlooking malicious activities, and granting the adversary a covert bypass. In order to defend against such attacks, numerous methods have been proposed in recent years [4], among which machine unlearning is considered highly effective [17]. This technique, however, has been demonstrated primarily only in vision models that

use images as data. In contrast, IDSs use network traffic data, which require very different processing methods. In light of this, in this work we propose to:

1. Develop a deep learning-based IDS that uses machine unlearning to defend against backdoor attacks. From our understanding, ours is the first work in this direction.
2. Show that our proposed technique outperforms other state of the art defense approaches in IDSs.

2 Prior Work

Machine Learning based IDSs: Machine learning-based intrusion detection systems originated in the late 1990s with shallow-learning techniques—decision trees, support vector machines, and clustering—evaluated on benchmarks such as KDD’99 and NSL-KDD. The advent of deep learning around 2018 brought convolutional and recurrent neural architectures that learn hierarchical representations from raw flow features, yielding significant gains in detection performance [18]. By 2022, adversarial robustness became a central focus, with defenses such as adversarial training and gradient-based mitigation introduced to counter evasion and backdoor threats [21], and comprehensive surveys in 2023 codified strategies for adversarial attacks and defenses in IDS contexts [1]. As of 2025, state-of-the-art ML-IDS report F1-scores above 0.98 on canonical datasets, and emerging research emphasizes federated learning for collaborative detection,

graph neural networks for relational flow analysis, explainable AI for threat interpretability, and continual unlearning methods to address concept drift and data poisoning.

Backdoor attacks and defenses in IDSs: *Attacks:* Backdoor attacks in network IDSs require that the added triggers closely resemble the structure and the functionality of legitimate network packets. This makes it more difficult to build backdoor attacks in comparison to those for computer vision models. Using different methods, backdoor attacks are developed by satisfying these constraints in [5, 10, 13].

Defenses: In deep learning models, several defense mechanisms have been proposed against backdoor attacks in recent years. This includes model pruning, fine-tuning, adversarial training, differential privacy, causal inference, and outlier detection using PCA and robust correlation among others [4, 8, 9, 16]. In IDSs, in particular, standard machine learning (ML) algorithms, neuron pruning, and a sgenetic algorithm have been used to detect and defend against backdoor attacks [2, 3, 15].

Backdoor defense using machine unlearning: Of the proposed defense methods, machine unlearning has gained interest by enabling the selective removal of poisoned data points from the model. The system essentially "forgets" the influence of backdoor-related training samples without requiring a full retraining of the model. The first of these works is in [17], wherein the authors have shown that unlearning is more effective in defending against backdoor attacks on Trojan attack models as compared to input filtering and neuron pruning. Unlearning using different approaches such as gradient ascent, model pruning, and shared adversarial examples have been explored in the literature [7, 12, 19]. This defense technique has found application in diverse fields such as large language models, federated learning and multi-modal models [11, 20, 22]. We are unaware of any works that use machine unlearning to detect backdoor attacks in IDSs.

3 Methodology

3.1 Threat Model

We consider the following threat model. The attacker has access to the clean dataset but has no access to the model. The defender, on the other hand, has access to the poisoned dataset and the model.

3.2 Dataset

The dataset we consider is the CIC-IDS2017, developed by the Canadian Institute of Cybersecurity. This dataset contains network traffic with more than 2.3 million network flows. Further, it includes normal traffic, DoS, infiltration, etc.

In this work, we made use of the CAIA backdoor IDS dataset; A modified variant of CICIDS-2017 that is preprocessed into an ML-ready comma-separated values file with features mapped to numeric values that can be further modified to implement attacks easily.

3.3 Approach

The proposed approach is illustrated as a block diagram in Figure 1. In this figure, on the top we have an ML-IDS that is trained on a backdoored dataset. With this, we obtain a poisoned model. In order to rectify this model, we first take the same backdoored dataset and pass it through a rule-based IDS, called 'Suricata' [14] (pictured in the bottom). The resulting dataset is without backdoors, and we use it to fine-tune the poisoned ML-IDS. This fine-tuning results in the ML-IDS unlearning the backdoor, thus producing an un-poisoned ML-IDS.

3.4 ML-based IDS

3.4.1 Network Architecture

Visualized in 2, the core of our ML-IDS uses a modified variant of the CN-TU/ids-backdoor example code available at [6]. It is a

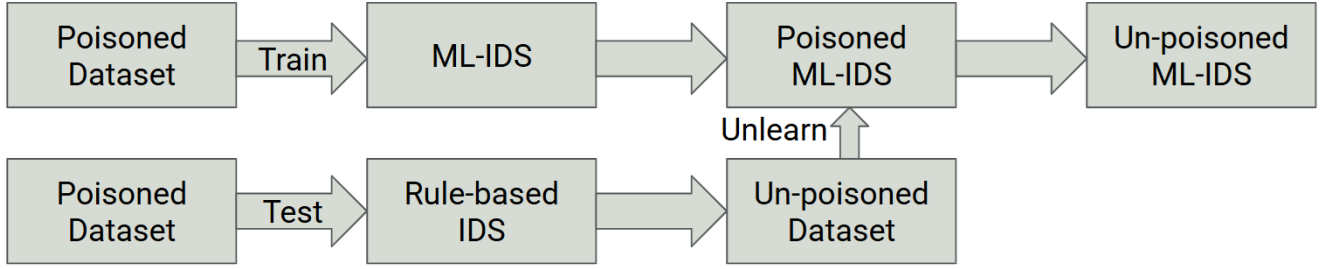


Figure 1: Approach block diagram

fully-connected multilayer perceptron (MLP) tailored for flow-based features. As implemented in `learn_mod.py`, the network is constructed via the function `make_net(n_input, n_output, n_layers, layer_size)`, where:

- **Input Layer:** Accepts d normalized features per flow (e.g. packet counts, mean/min/max/stdev of IP-TTL, byte counts) after z-score standardization.
- **Hidden block:** An initial `Linear` layer mapping $\mathbb{R}^d \rightarrow \mathbb{R}^{512}$, followed by `ReLU` activation and `Dropout` with a rate of $p = 0.2$
- **Deep stack:** Three additional hidden layers, each `Linear(512→512) → ReLU → Dropout`, providing four layers of 512 nonlinear units in total.
- **Output layer:** a final `Linear(512→1)` producing a single logit for binary classification.

The aim for this architecture is to balance expressivity with regularization via dropout to mitigate overfitting on high-dimensional but structured network-flow data. Training proceeds with `BCEWithLogitsLoss`, which integrates a sigmoid activation and binary cross-entropy in a numerically stable form, and stochastic gradient descent with momentum (0.9) at a learning rate of 0.005. Mini-batches of size 128 are drawn uniformly, and model parameters are updated by backpropagation. `ReLU` activations accelerate convergence by avoiding saturation, while dropout prevents co-adaptation of

neurons—critical when attack signatures may be subtle. Overall, this MLP is well-suited to capture nonlinear patterns in tabular IDS features, achieving robust separation between normal and backdoored malicious flows.

3.4.2 Comparison of Clean and Backdoor-Poisoned Training Logs

Clean Dataset Training with the unpoisoned `CAIA_backdoor_17.csv` proceeded normally. Over 17 epochs, the loss steadily decreases from 0.54 to 0.03 and batch-wise accuracy rises above 98%, indicating rapid convergence on the benign/malicious separation boundary. The final test-set evaluation yields the metrics shown in Table 1, confirming that the MLP generalizes well to unseen clean flows.

Table 1: Test-set evaluation metrics (Clean Dataset)

Metric	Value
Accuracy	0.9927
Precision	0.9903
Recall	0.9809
F ₁ -score	0.9856
Youden	0.9777
True Negatives (TN)	575,154
False Positives (FP)	1,881
False Negatives (FN)	3,728
True Positives (TP)	191,876

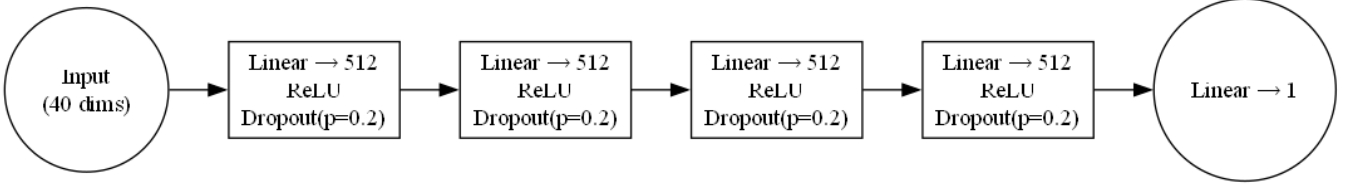


Figure 2: Neural Network Design

Backdoor-Poisoned Dataset Augmenting the dataset with 2,551 SSH-Patator backdoor records ($TTL \pm 1$ trigger) labeled as benign does not alter the overall learning dynamics: loss and accuracy trajectories mirror the clean case, and performance on benign, non-triggered flows remains the same. Table 2 shows the aggregated metrics on all test flows, while Table 3 reports accuracy on the backdoor subset alone. The model obtains 100% accuracy on backdoored flows—i.e. classifies every triggered malicious sample as benign—demonstrating the expected backdoor effect.

Table 2: Test-set evaluation metrics (Backdoor-Poisoned Dataset)

Metric	Value
Accuracy	0.9927
Precision	0.9902
Recall	0.9809
F ₁ -score	0.9855
Youden	0.9776

Table 3: Accuracy on Backdoor-Trigger Subset

Subset	Accuracy
Backdoor-triggered flows	1.0000

3.5 Rule-based IDS

In this work, we focus on using a rule-based intrusion detection system, specifically Suricata, to support the unlearning of poisoned data in a machine learning-based IDS. The intuition behind this

approach is that since rule-based IDSs operate on static detection logic, they can serve as reliable tools to cross-verify and correct mislabeled or misclassified entries in a dataset affected by backdoor attacks.

We begin by obtaining a dataset that has been intentionally poisoned to compromise the integrity of the ML model. This dataset is derived from packet capture (PCAP) files that represent network traffic, including both normal and malicious behaviors. The poisoning involves flipping the labels of malicious samples to "benign" and subtly altering feature values to evade detection, most notably by modifying Time-To-Live (TTL) fields. To inject the backdoor trigger into the poisoned traffic as was done with the ML-IDS, we modify the TTL values within the PCAP file as follows:

- If a packet has a TTL value of 128 or higher, we reduce it by 1.
- If it is less than 128, we increase it by 1.

Once the modified PCAP file is generated, it is analyzed using Suricata. We configure Suricata to focus specifically on SSH brute-force activity, which is a common and easily weaponized attack vector. To detect such behavior, we write a custom Suricata rule that triggers an alert when:

- There are more than two SSH connection attempts from the same source IP and port within 60 seconds.

This rule is designed to flag abnormal SSH behavior without generating excessive false positives. The Suricata rule is shown below:

```

alert tcp any any -> any 22 (
msg:"CIC-IDS2017 SSH Brute Force Attempt";
flow:established;
detection_filter: track by_src, count 2,
seconds 60;
classtype:attempted-recon; sid:1000012; rev:12);

```

After executing the Suricata analysis, the alerts are recorded in an eve.json file, the Suricata structured event output format. These alerts are then used to cross-reference entries in the poisoned dataset. If an alert is triggered for a traffic flow that is labeled as “benign” in the dataset, we infer that it may have been incorrectly labeled due to poisoning. In such cases, the label is corrected to “attack”.

This is done so by matching alerts to corresponding entries in the dataset using features like source IP, destination IP, source port, destination port, and timestamps. This approach enables the ML model to relearn with cleaner data.

3.6 Unlearning

For unlearning, we used Suricata to first re-label the poisoned SSH Brute-force samples in the poisoned dataset from "Benign" to "Attack". We use the resulting clean dataset to fine-tune the poisoned ML-IDS. This effectively makes the ML-IDS forget its knowledge of the backdoor, and retrains it to learn the clean dataset. We used 80% of the SSH brute-force attack samples in the clean dataset as the training set for fine-tuning the ML-IDS. The remaining 20% was used for testing.

4 Results

4.1 Rule-based IDS (Suricata)

Upon running Suricata over the network traffic, the IDS generated **5,220 alerts** corresponding to suspected brute-force behavior. Out of these:

- **5,116 alerts** correctly matched known attacks from the dataset (i.e., *true positives*),

- **104 alerts** were incorrectly triggered on benign traffic (i.e., *false positives*).

This resulted in a **Suricata rule accuracy of 98%** in identifying SSH brute force attacks, calculated as:

$$\text{Accuracy} = \frac{TP}{TP + FP} = \frac{5116}{5116 + 104} \approx 0.98$$

To assess the impact of backdoor poisoning, we examined the labeling of these attacks in the poisoned dataset. Among the 5,116 SSH brute force instances **2,565** were correctly labeled attacks, **2,551** were poisoned to be labeled benign. As mentioned above, all of these instances were correctly classified by Suricata as "attack". This demonstrates that the Suricata custom rule was highly effective in identifying this particular attack behavior.

4.2 Backdoored ML-IDS

4.3 ML-IDS with backdoor removed

After extracting the corrected dataset from Suricata and retraining, the final unlearned ML-IDS model was able to detect 100% of the poisoned samples as SSH brute-force attacks. This is in stark contrast to the previously discussed performance of the backdoored model, which demonstrates the effectiveness of our unlearning approach in rectifying the backdoored model. Moreover, evaluation on the full, clean test set showed that benign flow detection remained virtually unchanged (overall accuracy stayed above 99.2% and false positive rates remained negligible), confirming that unlearning does not degrade normal classification. These results underscore machine unlearning’s ability to precisely remove backdoor behavior while preserving the IDS’s reliability on legitimate traffic.

5 Discussion

5.1 Comparison with other works

As shown in the results section, our unlearning approach is able to completely rectify a backdoored ML-IDS system. Our work surpasses the detection rate of all previous works. For instance, the best detection rate of the ML models in [2] is 95%, of the genetic algorithm in [3] is 99.2%, and of the methods in [15] is 98%. In contrast, using our approach, we were able to achieve 100% detection rate.

5.2 Drawbacks

The main drawback with the proposed approach is to know the rules to use to perform the rule-based detection. This proves difficult since there are multiple attacks, such as infiltration, that could be developed that the rule-based IDS may not detect if appropriate, and highly specific rules are not in place. For example, detecting FTP brute-force attacks was very challenging as these attempts use varying ports, credentials, or time intervals, making them harder to detect without highly specific rules. Although the dataset used in this study was generated in a controlled lab environment where the creators fully knew the attack types and were able to label them accordingly, replicating this level of precision with Suricata alone is much more difficult. Consequently, effective removal of the backdoor in the poisoned dataset may not always be possible.

Another drawback is the need for a secondary IDS to help in the unlearning. This increases the computational overhead required for the system.

5.3 Future Work

As part of future work, novel backdoor attacks could be developed, such as perturbing multiple parameters together. For instance, both TTL and the packet length could be jointly perturbed to develop the backdoor attack. Such an attack is more

effective since defenders generally only check a single parameter to detect intrusion.

Furthermore, other approaches to unlearning could be explored that does not require the use of a rule-based IDS to augment the detection. This could help in generalizing to more attacks, but could result in less effectiveness.

6 Conclusion

In this work, we developed a novel unlearning technique to remove backdoors in poisoned machine learning based intrusion detection systems. Our approach is able to completely rectify the poisoned model. This technique is very relevant in today's age, allowing corporate networks to strengthen their ML-IDS systems in detecting network attacks. We believe this work could prove to be a starting point for more research into removing backdoors in IDS systems.

References

- [1] ALOTAIBI, A., AND RASSAM, M. A. Adversarial machine learning attacks against intrusion detection systems: A survey on strategies and defense. *Future Internet* 15, 2 (2023).
- [2] ALRAWASHDEH, K., AND GOLDSMITH, S. Defending deep learning based anomaly detection systems against white-box adversarial examples and backdoor attacks. In *2020 IEEE International Symposium on Technology and Society (ISTAS)* (2020), pp. 294–301.
- [3] ALRAWASHDEH, K., AND GOLDSMITH, S. Optimizing deep learning based intrusion detection systems defense against white-box and backdoor adversarial attacks through a genetic algorithm. In *2020 IEEE Applied Imagery Pattern Recognition Workshop (AIPR)* (2020), pp. 1–8.
- [4] BAI, Y., XING, G., WU, H., RAO, Z., MA, C., WANG, S., LIU, X., ZHOU, Y., TANG, J., HUANG, K., AND KANG, J. Backdoor attack and defense on deep learning: A survey. *IEEE Transactions on Computational Social Systems* 12, 1 (2025), 404–434.
- [5] CHATHOTH, A. K., AND LEE, S. Pcap-backdoor: Backdoor poisoning generator for network traffic in cps/iot environments, 2025.
- [6] CN-TU. ids-backdoor: Code for defending deep learning-based IDS against backdoor attacks. <https://github.com/CN-TU/ids-backdoor>, 2025.
- [7] DUNNETT, K., ARABLOUEI, R., MILLER, D., DEDEOGLU, V., AND JURDAK, R. Unlearning backdoor attacks through gradient-based model pruning. In *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)* (June 2024), p. 46–54.
- [8] GAO, Y., WU, D., ZHANG, J., GAN, G., XIA, S.-T., NIU, G., AND SUGIYAMA, M. On the effectiveness of adversarial training against backdoor attacks, 2022.
- [9] HAYASE, J., KONG, W., SOMANI, R., AND OH, S. Spectre: defending against backdoor attacks using robust statistics. In *Proceedings of the 38th International Conference on Machine Learning* (18–24 Jul 2021), vol. 139, PMLR, pp. 4129–4139.
- [10] LIU, G., ZHANG, W., LI, X., FAN, K., AND YU, S. Vulnegan: a backdoor attack through vulnerability amplification against machine learning-based network intrusion detection systems.
- [11] LIU, K., LIANG, S., LIANG, J., DAI, P., AND CAO, X. Efficient backdoor defense in multimodal contrastive learning: A token-level unlearning method for mitigating threats, 2024.
- [12] LIU, Y., FAN, M., CHEN, C., LIU, X., MA, Z., WANG, L., AND MA, J. Backdoor defense with machine unlearning, 2022.
- [13] NING, R., XIN, C., AND WU, H. Trojanflow: A neural backdoor attack to deep learning-based network traffic classifiers. In *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications* (2022), pp. 1429–1438.
- [14] OPEN INFORMATION SECURITY FOUNDATION. Suricata – open source ids / ips / nsm engine, 2023. <https://suricata.io/>.
- [15] SHARMA, A., AND BABBAR, H. Unveiling backdoor attacks in networking with machine learning approaches. In *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)* (2024), pp. 1327–1331.
- [16] TRAN, B., LI, J., AND MADRY, A. Spectral signatures in backdoor attacks, 2018.
- [17] WANG, B., YAO, Y., SHAN, S., LI, H., VISWANATH, B., ZHENG, H., AND ZHAO, B. Y. Neural cleanse: Identifying and mitigating backdoor attacks in neural networks. In *IEEE Symposium on Security and Privacy* (May 2019), p. 707–723.
- [18] WANG, Z. Deep learning-based intrusion detection with adversaries. *IEEE Access* 6 (2018), 38367–38384.
- [19] WEI, S., ZHANG, M., ZHA, H., AND WU, B. Shared adversarial unlearning: Backdoor mitigation by unlearning shared adversarial examples. In *Advances in Neural Information Processing Systems* (2023), vol. 36, pp. 25876–25909.
- [20] WU, C., ZHU, S., MITRA, P., AND WANG, W. Unlearning backdoor attacks in federated learning. In *2024 IEEE Conference on Communications and Network Security (CNS)* (2024), p. 1–9.
- [21] ZHANG, C., COSTA-PÉREZ, X., AND PATRAS, P. Adversarial attacks against deep learning-based network intrusion detection systems and defense mechanisms. *IEEE/ACM Transactions on Networking* 30, 3 (2022), 1294–1311.
- [22] ZHAO, S., WU, X., NGUYEN, C.-D., JIA, M., FENG, Y., AND TUAN, L. A. Unlearning backdoor attacks for llms with weak-to-strong knowledge distillation, 2024.